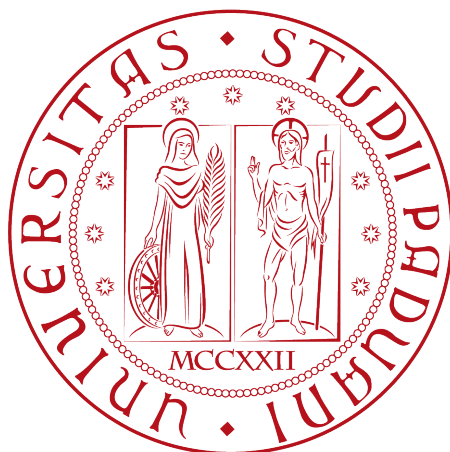


University of Padua

DEPARTMENT OF MATHEMATICS “TULLIO LEVI-CIVITA”

MASTER DEGREE IN COMPUTER SCIENCE



GeoMedia: A Geographic Framework for Location-Based Content Sharing

Master's Thesis

Supervisor

Prof. Claudio Enrico Palazzi

Co-supervisor

Dott. Lorenzo Perinello

Candidate

Alberto Morini

Student ID

2107783

ACADEMIC YEAR 2025-2026

Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT 2025: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [1]

Contents

1	Introduction	1
1.1	Overview	1
1.2	Social networks	2
2	Related works	3
2.1	Instagram and Snapchat	3
2.2	Google Maps	3
2.3	Strava	3
2.4	WallaMe	3
3	GeoMedia	4
3.1	Idea	4
3.2	Functionalities	4
4	Conclusions	5
4.1	Consuntivo finale	5
4.2	Raggiungimento degli obiettivi	5
4.3	Conoscenze acquisite	5
4.4	Valutazione personale	5
A	Appendice A	7
	Acronimi e abbreviazioni	8
	Glossario	9
	Bibliografia	10

List of Figures

List of Tables

List of Codes

Chapter 1

Introduction

1.1 Overview

Technology has always influenced society, since the first introduction to the personal computer to internet, changing the world year after year, till to make technology as a fundamental pillar of everyday of almost every human being. Over last decades, the investments and developing of technological devices has been increased exponentially, shaping the economical and social impact to adapt and adopt latest features and always new functionality and procedures. From personal life to workplaces, people are surrounded of software on computers and even more on mobiles, in fact, smartphones and tablets has been widely adopted till to put the development of applications “mobile firsts”.

The wider adoption of smartphones has reached almost three quarter of global population [13, 11], making users abandon traditional computer not just with web browsing but more especially with social media[10] and instant messages platform[12].

The strenghts of mobile devices can be divided onto the bare logic of any technology:

1. **Hardware:** the increase in computational performance[7], battery life[4], fast connectivity, and the integration of several built-in sensors[6] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[2], education[9], and, through motion-sensors or cameras, a coexistence with physical world, creating a more immersive and user experience.
2. **Software:** the development and presence of mobile applications has played a crucial role. In fact, platforms such as ‘Windows Phone’ were eventually abandoned due to the lack of applications [3, 14] compared to operating system like ‘iOS’ and ‘Android’.

There are many examples of this trend in modern society. For instance, when planning a trip or driving, smartphones assist users through apps like Google Maps or Waze. These apps don’t just provide directions—they also improve the experience with real-time traffic updates, road event notifications, and information about available parking. In this way, the surrounding environment is enriched with digital details. Another example is Google Lens, which can identify objects from a photograph, or Shazam that can recognize a song through the device’s microphone.

The technology integration has become an important key even in workplaces and public spaces, from the resturant that provide a digital menu by a tablet, to customers checking a shopping lists. Similary, many museums allowing visitors to scan QR codes to access audio-guided tours or unlock Augmented Reality functionalities that enhance the experience.

1.2 Social networks

Alongside the adoption of mobile technologies, social networks have emerged as a main role in the technology market, placing their position among the most used applications globally. These platforms have totally changed the way people communicate digitally, by providing real-time messaging capabilities in both one-to-one and one-to-many formats, while also allowing users to share any kind of multimedia, including photos, videos, audio recordings, and other files. Applications such as WhatsApp, Telegram, Instagram, and Facebook have attracted billions of users worldwide, becoming deeply embedded in daily communication practices across diverse social, professional, and public contexts.

By facilitating the instantaneous exchange of information and supporting multiple forms of interaction, social networks have not only expanded the scope of personal and professional communication but have also transformed the ways in which knowledge, ideas, and content are disseminated and consumed. These platforms are predominantly user-centric, leveraging personalized recommendation algorithms that tailor content according to users' interests, interactions, and browsing behavior. As a result, they foster engagement while facilitating the creation and maintenance of digital relationships across diverse communities. By bridging geographical distances and enabling individuals to connect around shared interests, social networks have contributed to the development of increasingly interconnected digital ecosystems. Beyond their communication functions, these platforms have evolved into important channels for entertainment, news consumption, content creation, and professional networking, reflecting their multifaceted role in contemporary society. Consequently, social media has redefined the landscape of digital connectivity, creating new opportunities for collaboration, engagement, and participation on a global scale.

Chapter 2

Related works

SOCIAL MEDIA

- snapchat - instagram - google maps / trip advisor - foursquare - jagat - auglinn - wallame??

2.1 Instagram and Snapchat

2.2 Google Maps

2.3 Strava

2.4 WallaMe

Chapter 3

GeoMedia

3.1 Idea

3.2 Functionalities

Chapter 4

Conclusions

- 4.1 Consuntivo finale
- 4.2 Raggiungimento degli obiettivi
- 4.3 Conoscenze acquisite
- 4.4 Valutazione personale

“Life is really simple, but we insist on making it complicated”

— Confucius

Ringraziamenti

Innanzitutto, vorrei esprimere la mia gratitudine al Prof. Claudio Enrico Palazzi, relatore della mia tesi, per l'aiuto e il sostegno fornitomi durante la stesura del lavoro.

Desidero ringraziare con affetto i miei genitori per il sostegno, il grande aiuto e per essermi stati vicini in ogni momento durante gli anni di studio.

Ho desiderio di ringraziare poi i miei amici per tutti i bellissimi anni passati insieme e le mille avventure vissute.

Padua, May 2026

Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronimi e abbreviazioni

API [Application Program Interface](#). [9](#)

UML [Unified Modeling Language](#). [9](#)

Glossario

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. [8](#)

UML in ingegneria del software *UML, Unified Modeling Language* (ing. linguaggio di modellazione unificato) è un linguaggio di modellazione e specifica basato sul paradigma object-oriented. L'*UML* svolge un'importantissima funzione di “lingua franca” nella comunità della progettazione e programmazione a oggetti. Gran parte della letteratura di settore usa tale linguaggio per descrivere soluzioni analitiche e progettuali in modo sintetico e comprensibile a un vasto pubblico. [8](#)

Bibliografia

Riferimenti bibliografici

- [5] Daniel T. Jones James P. Womack. *Lean Thinking, Second Editon*. Simon & Schuster, Inc., 2010.

Siti web consultati

- [1] Lorenzo Perinello. Alberto Morini Claudio Enrico Palazzi. *GoodIT 2025 - International Conference on Information Technology for Social Good*. 2025. URL: <https://goodit2025.org> (cit. on p. ii).
- [2] Shariq Aziz Butt et al. *Remote mobile health monitoring frameworks and mobile applications: Taxonomy, open challenges, motivation, and recommendations*. 2024. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0952197624003919> (cit. on p. 1).
- [3] Windows Central. *Windows Phone isn't dead, Part VI: App Gap? Microsoft has a platform for that*. 2016. URL: <https://www.windowscentral.com/windows-phone-isnt-dead-part-vi-app-gap> (cit. on p. 1).
- [4] Aoxia Chen and Pankaj K. Sen. *Advancement in battery technology: A state-of-the-art review*. 2016. URL: <https://ieeexplore.ieee.org/document/7731812> (cit. on p. 1).
- [6] Peter James Marcin Strackiewicz and Jukka-Pekka Onnela. *A systematic review of smartphone-based human activity recognition methods for health research*. 2021. URL: <https://www.nature.com/articles/s41746-021-00514-4> (cit. on p. 1).
- [7] Yuhao Zhu Matthew Halpern and Vijay Janapa Reddi. *Mobile CPU's rise to power: Quantifying the impact of generational mobile CPU design trends on performance, energy, and user satisfaction*. 2016. URL: <https://ieeexplore.ieee.org/document/7446054> (cit. on p. 1).
- [8] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *Mobile internet usage world-wide - statistics and facts*. 2023. URL: <https://www.statista.com/topics/779/mobile-internet/>.
- [9] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *The perspectives for mobile application use in media education*. 2023. URL: <https://www.tandfonline.com/doi/full/10.1080/10494820.2023.2186897> (cit. on p. 1).
- [10] Statista. *Device usage of Facebook users worldwide as of January 2022*. 2022. URL: <https://www.statista.com/statistics/377808/distribution-of-facebook-users-by-device/> (cit. on p. 1).

- [11] Statista. *Global smartphone sales to end users from 2007 to 2023*. 2023. URL: <https://www.statista.com/statistics/263437/global-smartphone-sales-to-end-users-since-2007/> (cit. on p. 1).
- [12] Statista. *Most popular global mobile messenger apps as of October 2025, based on number of monthly active users*. 2025. URL: <https://www.statista.com/statistics/258749/most-popular-global-mobile-messenger-apps/> (cit. on p. 1).
- [13] International Telecommunication Union. *International Telecommunication Union. 2023. Facts and Figures 2023 – Mobile Phone Ownership*. 2023. URL: <https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-mobile-phone-ownership/> (cit. on p. 1).
- [14] Computer World. *Windows Phone still has serious app gap despite Microsoft’s \$7.2 billion Nokia buyout*. 2013. URL: <https://www.computerworld.com/article/1500308/windows-phone-still-has-serious-app-gap-despite-microsoft-s-7-2-billion-nokia-buyo.html> (cit. on p. 1).